

NGUYEN GIA HUY

Python Backend Engineer

johnnynguyen882@gmail.com | 0948385149 | linkedin.com/in/huy8bit

PROFESSIONAL SUMMARY

Python backend engineer working in core banking at Vikki Digital Bank: develops and maintains the **Lending** and **Overdraft** products as **Python Smart Contracts** on Thought Machine Vault Core (interest accrual, penalty and late fees, repayment, and settlement) along with the integrations and test infrastructure around them. Nearly three years of backend and data experience, including AWS ETL pipelines for au Denki, a utility platform serving millions of users, where batch throughput rose by **150%**, and multi-tenant authentication for a fintech cloud.

TECHNICAL SKILLS

Python:	Python 3, Thought Machine Vault Core Smart Contracts, FastAPI
Core Banking Logic:	Lending, overdraft, interest accrual, penalty and late fees, repayment, settlement, scheduled events
Testing:	Simulation testing, automated test frameworks
Backend & APIs:	REST/JSON, gRPC, Kafka, event-driven services, Keycloak (RBAC)
Data:	SQL, PostgreSQL, Amazon Athena, Redis, MongoDB, OpenSearch
AWS:	Lambda, ECS, RDS, S3, EKS
Also used:	Golang

WORK EXPERIENCE

Python Backend Engineer, Core Banking (Middle Software Engineer) 03/2026 – Present
Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

Lending and Overdraft Products (Python Smart Contracts)

- Developed and maintained the **Lending** and **Overdraft** products of Vikki Digital Bank as **Python Smart Contracts** on Thought Machine Vault Core.
- Implemented **interest accrual** for both products in contract code, running on Vault Core's contract hooks and **scheduled events**.
- Implemented **penalty and late-fee calculation**, applying charges when payments fall overdue.
- Implemented **repayment** processing, where each calculation flows directly into customer balances on the core ledger.
- Implemented **settlement** logic that closes out the amounts owed under each product.

Integration and Testing

- Integrated Vault Core with external backend services that drive core banking operations and transaction workflows.
- Rebuilt the framework that **simulates contract behaviour** over time, moving it entirely from Gauge to Golang.
- Cut simulation test runs from **24 minutes to just over 50 seconds (20x faster)**, so contract changes can be verified quickly.

Data and Platform

- Moved account data faster: built a Golang migration system on ECS, Lambda, and RDS that brought each run from **2.5 hours to 27 minutes (5x)**.
- Deployed the contract platform on AWS, a new Vault Core version on EKS, RDS, ECR, and ELB, and automated its build, test, and deploy pipeline.

Software Engineer – Data Processing

04/2025 – 03/2026

Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Designed, built, and operated **ETL pipelines** on AWS Lambda, ECS, RDS, S3, and Athena, transforming and processing large-scale utility data.
- Delivered processed data to downstream analytics through Athena on S3 and to operational workloads backed by RDS.
- Increased batch processing throughput by **150%** by redesigning pipelines, running work in parallel, and optimizing data access patterns.
- Reworked the ingestion and indexing side of search while moving it from **Elasticsearch to OpenSearch**, for more stable, consistent, and faster search and analytics queries.
- Supported the pipelines in production: infrastructure in Terraform and AWS SAM, plus health, failure, and anomaly monitoring in CloudWatch and DataDog.

Software Engineer

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Gave **20+ customer organizations** their own isolated sign-in through multi-tenant **authentication** in Keycloak.
- Implemented role-based **authorization** on top of it, ensuring tenant isolation between organizations.
- Put **100+ microservices** behind **Kong Gateway** access control at both the service and the network level, and operated the gateway.
- Operated VNPAY Cloud's OpenStack private cloud: **50+ production Kubernetes clusters** with Cluster API-based provisioning, scaling, and zero-downtime upgrades.
- Took about **80%** of data center provisioning and infrastructure configuration off manual processes through automation.

Blockchain Developer

06/2023 – 09/2023

EsoiLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Developed **15+** DeFi and NFT smart contracts in Solidity and Move and took part in contract reviews.
- Processed on-chain events in event-driven services and synced the results into **off-chain databases and APIs**.
- Integrated them with **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui), keeping transaction handling consistent and data intact.

SELECTED PROJECTS

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on **LiveKit**.
- Integrated **OpenAI models** for live transcription, voice translation, and meeting summarization.
- Designed the **real-time translation pipeline** and optimized conversational context handling to improve translation accuracy and coherence.

DattingQ | Software Engineer

Dating platform

- Built a gRPC-Gateway, Kafka, MongoDB, and Redis backend serving up to **1.5K requests/second** at **80–100 ms**.
- Implemented multi-cluster failover and observability (OpenTelemetry, Prometheus, Grafana, ELK, ClickHouse) for **99.95% uptime**.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera